

Mr Amir Hossein Dakhili

PhD Candidate

Neuroscience of Addiction & Mental Health Program, [HBMRC](#)

School of Behavioural & Health Sciences, Faculty of Health Science, Australian Catholic University

Daniel Mannix Building, Fitzroy, VIC 3065

Email: amirhossein.dakhili@myacu.edu.au

[Google Scholar](#) | [ResearchGate](#) | [Linkedin](#) | [GitHub](#)

RESEARCH BACKGROUND

I am a PhD Candidate at the Australian Catholic University, focused on characterizing and modulating the neurocircuitry of cannabis cue-induced craving using fMRI and real-time neurofeedback. My research aims to uncover fundamental neurobiological mechanisms of addiction, including Cannabis and Methamphetamine Use Disorders. To date, in the final year of my PhD, I have attracted a grant value of 112,000 A\$ from all awards and scholarships. I have been involved in 9 projects to date across distinct mental health disorders and using multi-modal imaging and neuromodulation tools, such as fMRI neurofeedback. I have published six papers in journals such as *Neuroscience & Biobehavioral Reviews* (IF = 7.9), *Addictive Behaviors* (IF = 3.6), *Drug and Alcohol Dependence* (IF = 3.6), *Journal of Neuroradiology* (IF = 3.3), and *Scientific Reports* (IF = 3.9; H-index: 2, Citations: 51), including two as lead author. I currently have additional manuscripts under review, including two as lead author.

EDUCATION

PhD Candidate, Neuroscience of Addiction & Mental Health Program March 2024 - 2027
HBMRC, Australian Catholic University, Melbourne, Australia

- ◊ Thesis: Characterizing and modulating the neurocircuitry of cannabis cue-induced craving using functional MRI and real-time neurofeedback
- ◊ Principal Supervisor: Prof. Valentina Lorenzetti.
- ◊ Co-Supervisors: Dr. Chao Suo, Dr. Govinda Poudel.

M.Sc. Medical Imaging Technology Sep 2019 - Sep 2022
Iran University of Medical Sciences, Tehran, Iran

- ◊ Thesis: Investigating the influence of pleasant and unpleasant odors on functional connectivity among the areas participating in decision making, using Functional Magnetic Resonance Imaging (fMRI).
- ◊ Thesis Supervisor: Dr. Arash Zare Sadeghi.
- ◊ Thesis Advisors: Dr. Ali Shakeri-Zadeh, Dr. Malake Malekzadeh.
- ◊ GPA: 17.28/20

B.Sc. Radiology Technology Sep 2014 - Jun 2018
Tabriz University of Medical Sciences, Tabriz, Iran

- ◊ GPA: 16.61/20

SKILLS

Medical Image Modalities

- Hands-on experience with MRI, CT Scan, and Radiography during apprenticeships at hospitals affiliated with Iran University of Medical Sciences and Tabriz University of Medical Sciences.

Neuroimaging:

- Hands-on experience with fMRI, including Proficiency in analyzing fMRI data using neuroimaging analysis tools (eg., CONN, FSL, and SPM).
- Proficiency in analyzing neuroimaging data using various analysis tools.
- Expertise in fMRI preprocessing techniques using fMRIPrep to ensure data quality and reliability.
- Experience with Turbo_BrainVoyager software for real-time neurofeedback analysis.
- Skilled in fMRI experiment design, including task design using MATLAB.

Data Handling, Integration, & Analysis:

- Basic knowledge of integrating diverse datasets, such as clinical assessments with neuroimaging data, for holistic analysis.
- Adept at employing statistical methods for data analysis and drawing meaningful insights.
- Understanding of models predicting disease progression based on varied input data.
- Competence in utilizing tools to visualize complex datasets in an accessible manner.
- Basic proficiency in Python and MATLAB for data processing, with a foundational knowledge of R.

Psychological Assessment:

- Familiarity with administration of a range of psychological and behavioral assessments, including Wechsler Abbreviated Scale of Intelligence (WASI), Structured Clinical Interview for DSM-5 for Cannabis Use Disorder (SCID for CUD), and Timeline Followback (TLFB).

Operating system:

- Proficiency in Linux (Ubuntu) and Windows.

RESEARCH EXPERIENCE

Research Assistant

May 2026 – Present

Neuroscience of Addiction & Mental Health Program, Australian Catholic University, Melbourne, Australia

- ◇ Conduct evidence synthesis and support the program's neuroimaging studies and operations.

Research Assistant

Jan 2025 – Present

Melbourne University, Melbourne, Australia

- ◇ Assist with technical setup and neurofeedback protocols for emotion regulation in depression, contributing to the study design and data collection.
- ◇ Supervisors: Dr. Elena Pozzi, Prof. Valentina Lorenzetti, Dr. Chao Suo

Research Assistant

Apr 2020 – Mar 2022

Neuroimaging Analysis Group (NIAG), Tehran University of Medical Sciences, Tehran, Iran

- ◇ Contributed in designing the methodology, implementing formal fMRI analysis, data management and writing the manuscript of fMRI Drug Cue-Reactivity(FDCR) study
- ◇ Supervisors: Dr. Hamed Ekhtiari, Dr. Amir Hossein Batouli

Research Assistant

Mar 2020 – Nov 2021

Virtual Collaborative Networks (ViCoNs), Online

- ◇ Organized Neuroimaging team
- ◇ Held congress panel on Methodological and Conceptual Aspects of fMRI Cue Reactivity Studies in 14th International Addiction Science Congress (ASC), December 2020.
- ◇ Supervisor: Dr. Hamed Ekhtiari

PUBLICATIONS

In Preparation

1. **L Greenwood**, G Abbott, J G Bartschi, S Dunsford, I Goodwin, **A Dakhili**, A Paloubis, M Quinones Valera, E McTavish, A Verdejo-Garcia, J Cousijn, N Solowij, V Lorenzetti. The neuro-circuitry of cannabis cue-induced craving in Cannabis Use Disorder.

Under Review

2. **A Dakhili**, E Murphy, S Ganesan, A Zalesky, R Glarin, H Thomson, A Paloubis, G Poudel, C Suo, V Lorenzetti. Investigating upregulation of craving in Cannabis Use Disorder using real-time fMRI neurofeedback at ultra-high field MRI. Under review at *Human Brain Mapping*.
3. **A Dakhili**, E Murphy, S Ganesan, A Zalesky, R Glarin, H Thomson, A Paloubis, G Poudel, C Suo, V Lorenzetti. An fMRI-neurofeedback system to train the reward neurocircuitry in ultra-high field MRI. Under review at *Journal of Neural Engineering*.
4. V Lorenzetti, H Thomson, I Goodwin, **A Dakhili**, N Solowij, M Yucel, E Christensen. A new, co-designed repository of cannabis images from street cannabis: the CANNA dataset. [OSF Link](#)

Accepted

5. I Goodwin, D Oliver, E Chesney, A Gaillard, S Wang, A Wong Koo, K Petrilli, M Skumlien, **A Dakhili**, S Wigroth, S Obreshkova, S Yusufzai, R Vandrey, M Krausz, T Freeman, P McGuire, J Strang, V Lorenzetti, A Englund (2026). The subjective effects of Δ^9 -tetrahydrocannabinol: A systematic review and dose-response meta-regression. *Neuroscience & Biobehavioral Reviews*, 106795.
[DOI: 10.1016/j.neubiorev.2026.106795](#) **Impact Factor: 7.9 (Q1)**.
6. E Murphy, G Poudel, **A Dakhili**, S Ganesan, A Zalesky, R Glarin, H Thomson, S Lin, A Paloubis, S Kamboj, T Sava, B Moffat, V Manning, C Suo, V Lorenzetti (2026). Exploring fMRI-neurofeedback to reduce craving-related brain activity in people with cannabis use disorder: A feasibility study. *Addictive Behaviors*, 181, 108741.
[DOI: 10.1016/j.addbeh.2026.108741](#) **Impact Factor: 3.6 (Q1)**.
7. **A Dakhili**, S.K Kamrava A Zare-Sadeghi (2026). Smelling decides: fMRI evidence reveals the influence of pleasant and unpleasant odors on risky decision-making. *Journal of Neuroradiology*, 53, 101529.
[DOI: 10.1016/j.neurad.2026.101529](#). **Impact Factor: 3.3 (Q1)**.
8. **A Dakhili**, A Sangchooli, S Jafakesh, M Zare-Bidoki, G Soleimani, S A H Batouli, M A Oghabian, K Kazemi, A Faghiri, H Ekhtiari (2022). Cue-induced craving and negative emotion disrupt response inhibition in methamphetamine use disorder: Behavioral and fMRI results from a mixed Go/No-Go task. *Drug and Alcohol Dependence*, 233, 109353.
[DOI: 10.1016/j.drugalcdep.2022.109353](#) **Impact Factor: 3.9 (Q1)**.
9. E Murphy, **A Dakhili**, S Ganesan, A Zalesky, R Glarin, H Thomson, A Paloubis, B Moffat, S Kamboj, G Poudel, C Suo, V Lorenzetti (2025). CannChange: A protocol for a feasibility study using fMRI-based neurofeedback to change the neurobiology of craving in Cannabis Use Disorder. *British Medical Journal open*, 31;15(8):e105854.
[DOI: 10.1136/bmjopen-2025-105854](#) **Impact Factor: 2.4 (Q1)**.
10. S Jafakesh, A Sangchooli, A Aarabi, M S Helfroush, **A Dakhili**, M A Oghabian, K Kazemi, H Ekhtiari (2021). Temporally Dynamic Brain Activity During Drug Cue Reactivity and Response Inhibition in Methamphetamine Use Disorder. *Scientific Reports*, 12(1), 1-18.
[DOI: 10.1038/s41598-022-05619-8](#) **Impact Factor: 3.8 (Q1)**.

PRESENTATIONS

SYMPOSIUM

V Lorenzetti, C Suo, R Glarin, S Ganesan, E Murphy, **A Dakhili**, E Pozzi (Brisbane, Australia, June 2025).

fMRI-based neurofeedback: a tool to probe & target neurocircuitry across health & disease (Brisbane, Australia, June, 2025)

Organization for Human Brain Mapping (OHBM) International Conference.

ORAL PRESENTATION

A Dakhili, E Murphy, S Ganesan, G Poudel, A Paloubis, S Lin, B Moffat, A Zalesky, R Glarin, C Suo, V Lorenzetti (Melbourne, Australia, November 2025).

Probing the Craving Neurocircuitry in Cannabis Use Disorder Using Real-Time fMRI Neurofeedback. *Australasian Cognitive Neuroscience Society (ACNS) Conference.*

A Dakhili, H Ekhtiari, A Sangchooli, H Mohaddes Ardebili, A Khojasteh, S Jafakesh (Online, October 2020).

Methodological and Conceptual Aspects of fMRI Cue Reactivity Studies: Example of a Mixed Go-No Go Drug Cue-Reactivity Study.

14th International Addiction Science Congress (ASC).

POSTER

A Dakhili, E Murphy, S Ganesan, G Poudel, A Paloubis, S Lin, B Moffat, A Zalesky, R Glarin, C Suo, V Lorenzetti (Canberra, Australia, November 2025).

Probing the Craving Neurocircuitry in Cannabis Use Disorder Using Real-Time fMRI Neurofeedback. *Australasian Society of Psychophysiology (ASP) Conference.*

A Dakhili, E Murphy, S Ganesan, G Poudel, A Paloubis, S Lin, B Moffat, A Zalesky, V Manning, H. Valerie Curran, A Honnedevassthana Arun, C Suo, V Lorenzetti (Melbourne, Australia, November, 2025). Mapping & Modulating the Activity of the Craving Neurocircuitry in Cannabis Addiction: A Real-time fMRI Neurofeedback Setup.

OHBM Australian Chapter Meeting and Maths in Brain.

A Dakhili, E Murphy, S Ganesan, G Poudel, A Paloubis, S Lin, B Moffat, A Zalesky, V Manning, H. Valerie Curran, A Honnedevassthana Arun, C Suo, V Lorenzetti (Melbourne, Australia, November, 2025). Mapping & Modulating the Activity of the Craving Neurocircuitry in Cannabis Addiction: A Real-time fMRI Neurofeedback Setup.

OHBM Australian Chapter Meeting and Maths in Brain.

A Dakhili, E Murphy, S Ganesan, G Poudel, A Paloubis, S Lin, B Moffat, A Zalesky, R Glarin, C Suo, V Lorenzetti (Brisbane, Australia, June, 2025). A setup for real-time fMRI Neurofeedback for Cannabis Craving: A Pilot Study

OHBM International Conference.

A Dakhili, E Murphy, S Ganesan, G Poudel, A Paloubis, S Lin, B Moffat, A Zalesky, V Manning, H. Valerie Curran, A Honnedevassthana Arun, C Suo, V Lorenzetti (Melbourne, Australia, November, 2024). Mapping & Modulating the Activity of the Craving Neurocircuitry in Cannabis Addiction: A Real-time fMRI Neurofeedback Setup.

Organization for Human Brain Mapping (OHBM) Australian Chapter Meeting and Maths in Brain.

A Dakhili, I Pesianian (Tehran, Iran, May 2017). CT Scan cancer risks.

33rd Iranian Congress of Radiology.

TEACHING EXPERIENCE

Guest Speaker – Biomedical Science Seminar Sep 2025
Australian Catholic University

- ◇ Invited by faculty to share insights on research methods and academic pathways with final-year Bachelor of Biomedical Science students.

Guest Lecturer – Psychology Honours Seminar Aug 2025
Australian Catholic University

- ◇ Delivered a lecture on fMRI and neurofeedback applications in addiction research to Psychology Honours students.

Workshop Tutor Apr 2021
Online Workshop on fMRI Analysis Tools

Teaching Assistant Sep 2020 – Dec 2020
Fundamentals of Cognitive Neuroscience International Course, Online

- ◇ Assisted Dr. Hamed Ekhtiari, an investigator at the Laureate Institute for Brain Research (LIBR), in test administration, curriculum development, and grading assignments. Additionally, provided students with relevant course materials and literature.

AWARDS AND SCHOLARSHIPS

- Australian Catholic University Research Travel Grant 2025: Awarded one of ten \$500 competitive grants to support research-related travel.
- Best 2024 PhD presentation Award, Healthy Brain and Mind Research Center conference Day, Australian Catholic University 2024.
- International Society for Biomedical Research on Alcoholism (ISBRA) Conference Scholarship 2024: Awarded \$500 to support attendance at the ISBRA conference.
- Research Training Program (RTP) Scholarship 2024: Awarded a competitive scholarship of \$37,000 per annum from the Australian Government to undertake PhD research.
- 14th International Addiction Science Congress (ASC) 2020: EZAMA Addiction Medicine Diploma of Honor
- Iran national university entrance for M.Sc. 2019: Ranked 15 among more than 2000 candidates in university admission national exam to M.Sc. Medical Imaging Technology
- Iran national university entrance for B.Sc. 2014: Ranked top 1% candidates in university admission national exam to B.Sc. Radiology Technology

COURSES AND CERTIFICATES

- Applied Suicide Interventions Training (ASIST) May 2024
- Positive psychology by Pennsylvania University, Coursera Apr 2022
- Artificial Intelligence by DeepLearning.AI Aug 2023
- Positive psychology by Pennsylvania University, Coursera Apr 2022
- Python Programming by Michigan University, Coursera Aug 2021
- Computational Neuroscience Summer School by Neuromatch Academy Jul 2021
- Cognitive Neuroscience by Dr. Hamed Ekhtiari, Online Nov 2020
- Basic and Advanced Neuroscience by Sharif University, Online Oct 2020

OTHER ACTIVITIES

Webinar Host

June 2026

Organization for Human Brain Mapping (OHBM), Australian Chapter

- ◇ Hosted the webinar “Bayesian Statistics in Neuroimaging” for the OHBM Australian Chapter, attended by 50 participants from across Australia.

Student Representative

Feb 2026 – Dec 2027

Organization for Human Brain Mapping (OHBM), Australian Chapter

- ◇ Represent student members within the OHBM Australian Chapter.
- ◇ Contribute to student-focused initiatives, outreach activities, and conference-related events.

Mentor

Feb 2026 - Present

OHBM, Student and Postdoc Special Interest Group (SP-SIG)

- ◇ Serve as a mentor within the global SP-SIG community, providing guidance on professional development support to international early-career researchers.

Founder and Instructor

Dec 2015 – Jun 2018

Rise and Shine English Club, Tabriz University of Medical Sciences, Tabriz, Iran

- ◇ Directed a team comprising 50 language professionals, including the managing directors and editorial board of the Rise and Shine club.
- ◇ Delivered English instruction to over 200 students from Tabriz Medical University as a second language.
- ◇ Published 9 issues of an English magazine for teaching and learning English, spanning three levels: basic, intermediate, and advanced.
- ◇ Honored with the Student Affairs Center Award for outstanding club performance in 2018, standing out amongst 20+ university clubs.

REFERENCES

- Dr. Valentina Lorenzetti, Professor, Australian Catholic University, Melbourne, Australia
PhD Principal supervisor
Email: Valentina.lorenzetti@acu.edu.au
- Dr. Chao Suo, Research Fellow, Monash University, Melbourne, Australia
PhD Co-supervisor
Email: chao.suo@monash.edu
- Dr. Govinda Poudel, Research Fellow, Monash University, Melbourne, Australia
PhD Co-supervisor
Email: govinda.poudel@acu.edu.au
- Dr. Arash Zare Sadeghi, Assistant Professor, Iran University of Medical Sciences, Tehran, Iran
Masters supervisor
Email: zare.a@iums.ac.ir
- Dr. Hamed Ekhtiari, Affiliate Investigator, Laureate Institute for Brain Research, Tulsa, OK
Research supervisor
Email: hekhtiari@laureateinstitute.org

- Dr. Ali Shakeri-Zadeh, Assistant Professor, Johns Hopkins University, Baltimore, Maryland
Masters Co-supervisor
Email: ashaker3@jhu.edu